Weekly Student Progress Performa

Team Id	CS26084	**Week no / Date**	Week 1 20 July 2026
Project Title	SentinelCore: A Network Detection and Incident Response Platform		
Members Present	Om Rashiya (24cs084), Yash Patel (24cs082)		
Mentor	Akshita Kadam		

WEEK 1 • Problem Definition and Product Direction

Section	Details to be filled by the team
Planned Tasks (from last week)	Define the SGP problem statement, identify target users, study existing home-network security tools, and establish the project scope.
Tasks Completed This Week	Finalized SentinelCore as a self-hosted network detection and incident-response platform. Defined the primary workflow: discover devices, collect network alerts, correlate suspicious activity, investigate incidents, contain threats, and generate evidence reports.
Commits Made	Total: 0 commits (research and planning week). Repository work had not started; requirements and module boundaries were documented before implementation.
Issues Created / Closed	Created: Need a security platform understandable to non-specialist home/small-network users. Closed: Agreed on a modular, local-first platform rather than a cloud-only monitoring service.
Milestone Status	~5% complete — on track. Problem definition, audience, objectives, and success criteria approved by the team.
Blockers / Challenges Faced	The initial scope was too broad. We reduced it to a staged pipeline with clearly separated discovery, detection, investigation, response, and reporting capabilities.
Plan for Next Week	Prepare the threat model, compare IDS technologies, and select the backend, database, queue, and frontend stack.

Research and Technical Study

Research objective. Define the SGP problem statement, identify target users, study existing home-network security tools, and establish the project scope.

Study completed. Compared Suricata, Snort, and Zeek for visibility, rule support, and structured event output. Suricata was selected because EVE JSON integrates cleanly with an application pipeline while retaining mature signature-based detection. The incident-response lifecycle was used to shape the product flow.

Detection approach decision

Criterion	Options considered	SentinelCore decision
Network visibility	Suricata / Snort / Zeek	Suricata sensor
Structured output	Alert and event formats	EVE JSON pipeline
Operator workflow	Separate tools or one flow	Discover → detect → investigate → respond → report
Deployment	Cloud-only or locally managed	Self-hosted platform

Decision carried into the project

Finalized SentinelCore as a self-hosted network detection and incident-response platform. Defined the primary workflow: discover devices, collect network alerts, correlate suspicious activity, investigate incidents, contain threats, and generate evidence reports.

Implementation Record and Website Evidence

Problem statement

Home and small-network users need one understandable place to see devices, detections, investigations, response actions, and evidence. The project was therefore framed as a complete operational workflow rather than a collection of disconnected security tools.

Intended workflow

The agreed sequence was asset discovery → network detection → event search → correlation → incident investigation → guarded containment → reporting. This sequence later became the product navigation and the module order used throughout the repository.

Product boundary

The platform would be locally deployable and explainable. Detection findings would remain traceable to a Suricata signature or a readable correlation rule; response and reporting would follow only after evidence was visible to an operator.

User-facing direction

The product introduction had to describe network defence in plain language. The identity and introductory visual shown in the evidence capture established the name SentinelCore and the SecureHome presentation used by the website.



Figure W1.1 — Project-owned SentinelCore interface mapped specifically to Week 1.

Evidence interpretation. The evidence screen presents the SentinelCore/SecureHome identity and the visual theme later carried into the public and authenticated website. Its visible controls, labels, status fields, and navigation provide direct evidence for the work described on this page.

Repository and Architecture Evidence

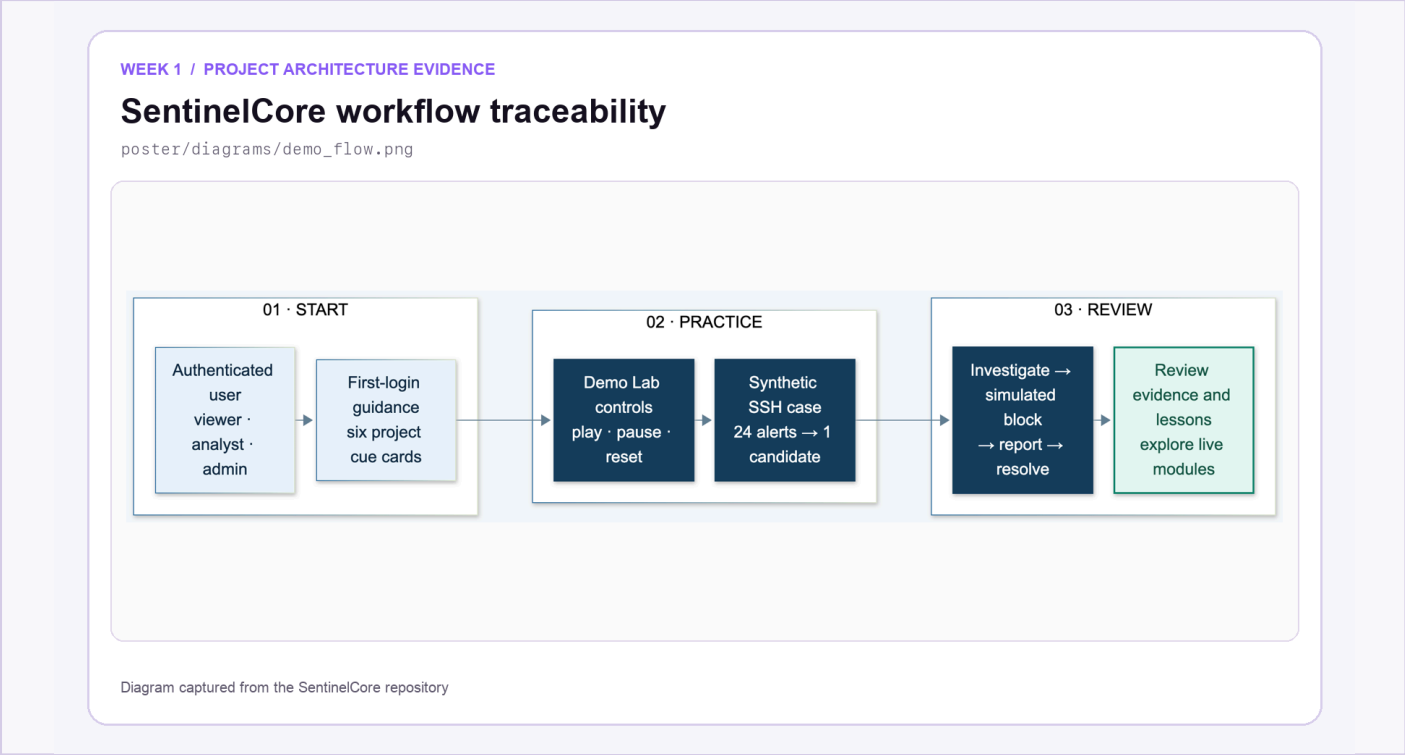


Figure W1.2 — SentinelCore workflow traceability. This is a separate internal artifact, not a copy of the website image.

Repository files used for traceability

- README.md — What it is and module workflow
- frontend/public/media/securehome-intro.mp4
- frontend/src/pages/LandingPage.jsx

Verification

Scope and workflow were checked against the final repository module list so the week records only capabilities that SentinelCore actually contains.

Traceability conclusion. SentinelCore workflow traceability directly supports the Problem Definition and Product Direction milestone. Scope and workflow were checked against the final repository module list so the week records only capabilities that SentinelCore actually contains. The artifact, source paths, implementation record, and website evidence therefore describe the same weekly scope.

Weekly Closure and Faculty Assessment

Completed outcome

Finalized SentinelCore as a self-hosted network detection and incident-response platform. Defined the primary workflow: discover devices, collect network alerts, correlate suspicious activity, investigate incidents, contain threats, and generate evidence reports.

Commit record

Total: 0 commits (research and planning week). Repository work had not started; requirements and module boundaries were documented before implementation.

Issue and resolution

Created: Need a security platform understandable to non-specialist home/small-network users. Closed: Agreed on a modular, local-first platform rather than a cloud-only monitoring service.

Milestone position

~5% complete — on track. Problem definition, audience, objectives, and success criteria approved by the team.

Challenge handled

The initial scope was too broad. We reduced it to a staged pipeline with clearly separated discovery, detection, investigation, response, and reporting capabilities.

Next-week hand-off

Prepare the threat model, compare IDS technologies, and select the backend, database, queue, and frontend stack.

Repository reference

https://github.com/rashiyaom/SentinelCore

Assessed By (Faculty)	Marks Awarded (5 marks)	Faculty Remarks